

Date: 2026/3/30

Computer Networks

Topics :

- Data Link Layer Review
- Data errors at Data link layer
- Error Detection
- Error Detection Methods
- Flow control of Data

Data Link Layer - Assignment 4

Present by :

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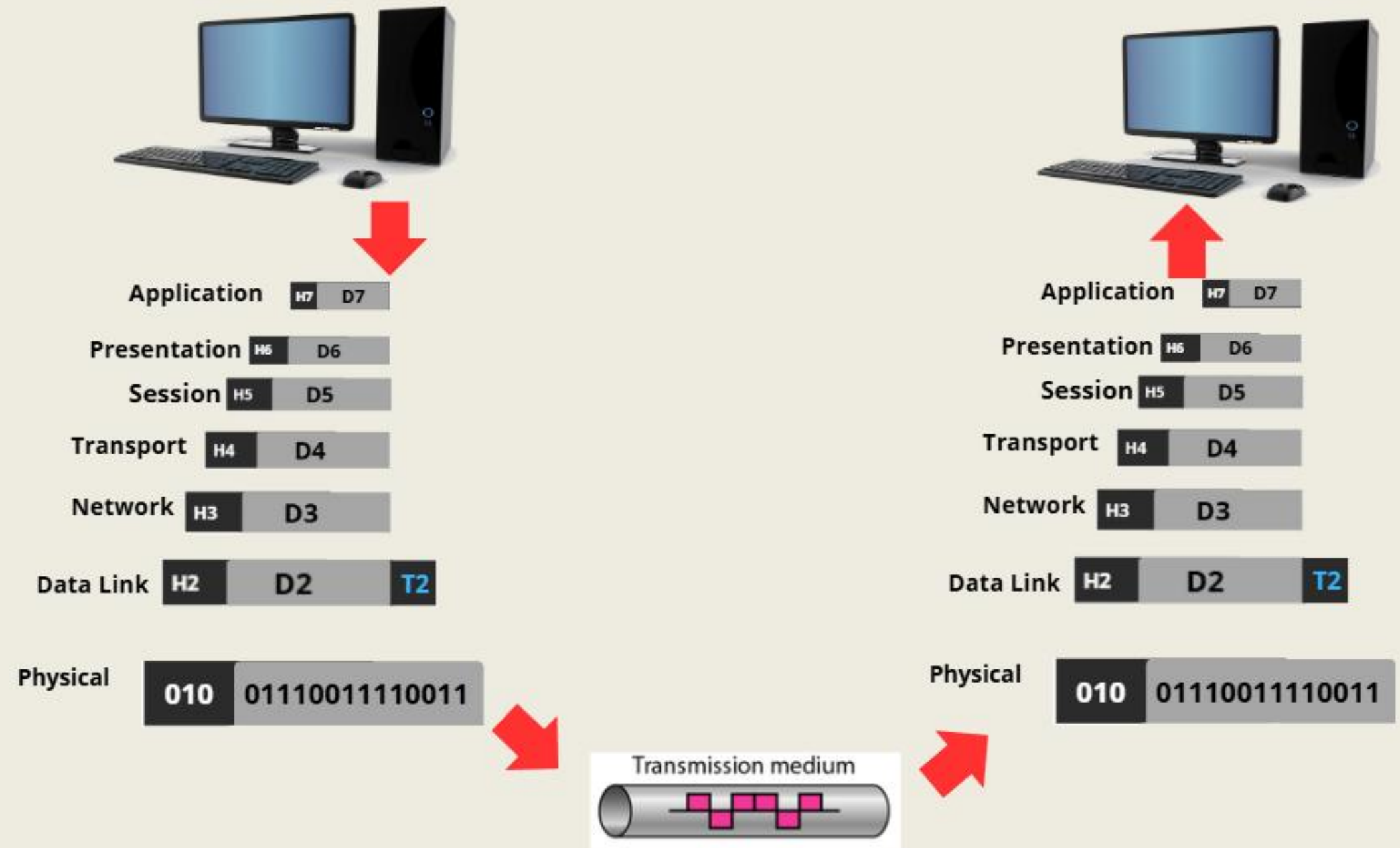
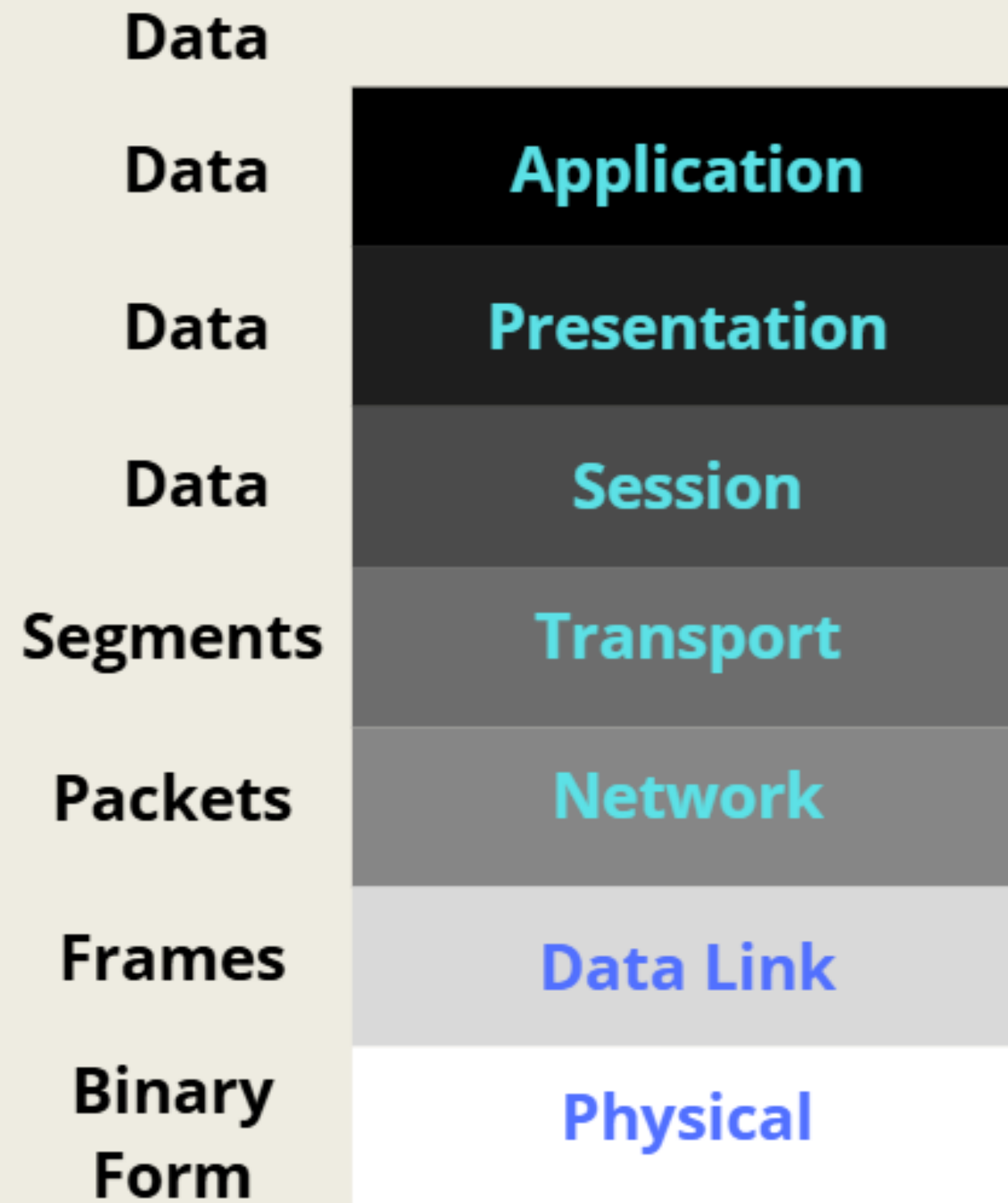
General Review

Network used Layers and Layered Models for Network Architecture.

The OSI model was theoretical and complex in implementation and setup.

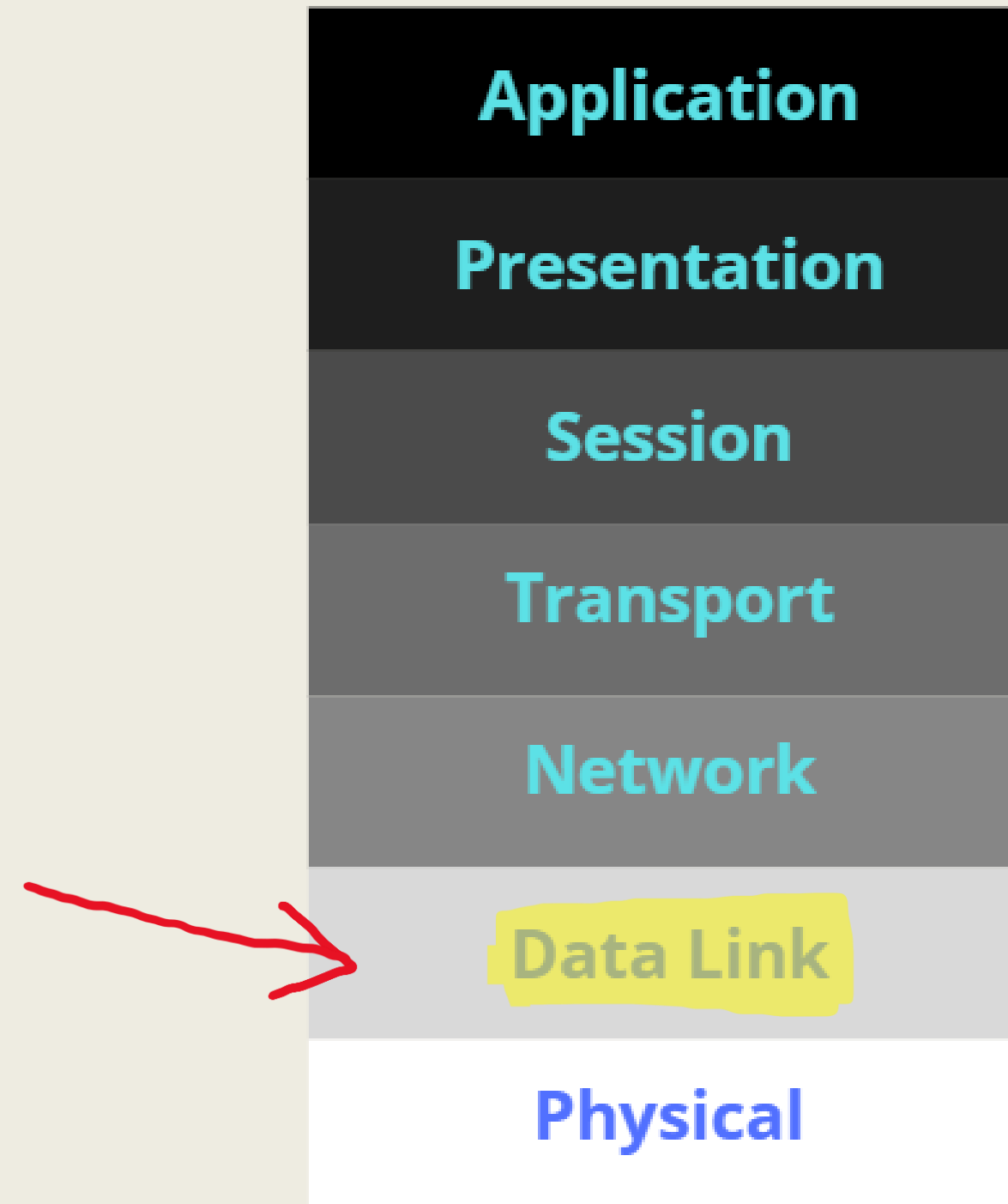
There are 7 Layers for OSI model , We will showing and discussing it.

General Review



General Review

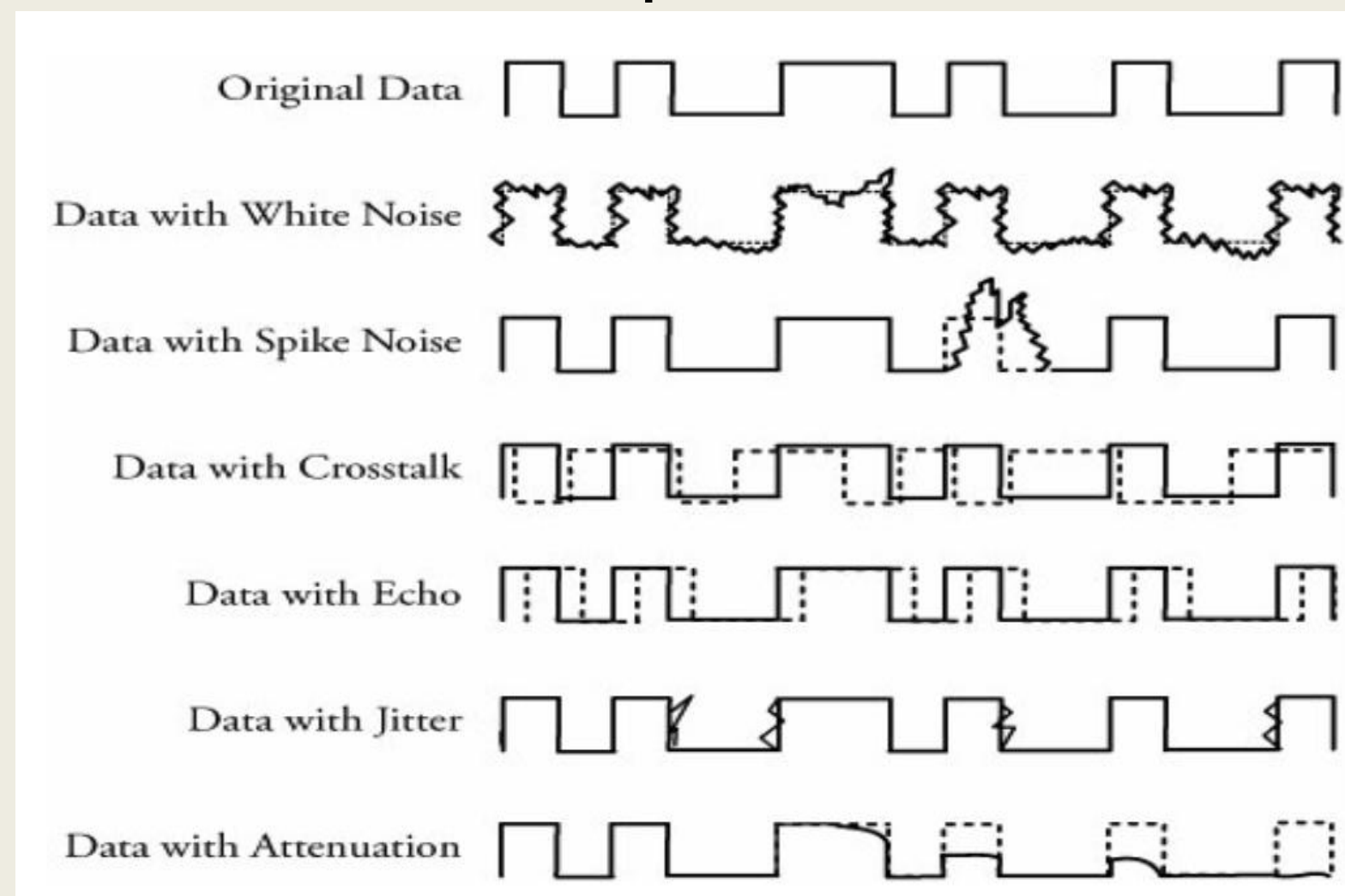
In this presentation we will represented and focused about Data Link layer (Data errors in this layer – Error detection method and Flow control)



Data Errors

In the data link layer, errors may happen and effect the data that affect its quality and accessibility, and may even prevent it from reaching its destination.

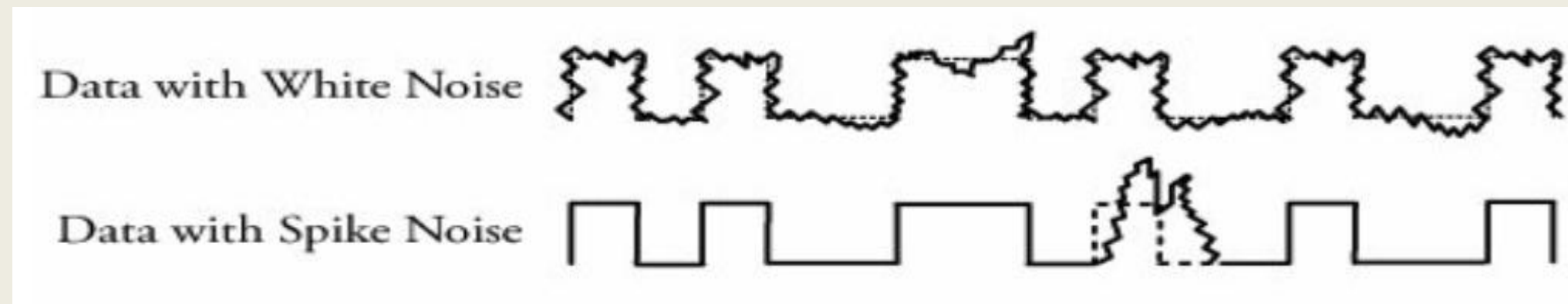
Examples :



Data Errors

White Noise : is **continuous** and is dependent on the temperature of the medium , Can be removed by passing the noise using filters.

Spike Noise : is **not-continuous** and maybe destroyed important part from data , And maybe cannot be recovered.



Data Errors

Cross Talk : Interference between two or more active links , can be electrical or electromagnetic.

Echo Noise : Reflection of transmitted signal

Jitter : Irregular timing caused errors when signal rises to 1 or falls to 0 .

Attenuation : Loss of signal strength by internal or external factors , Can be solved using repeaters and amplifiers .

Data Errors

Data with Crosstalk



Data with Echo



Data with Jitter



Data with Attenuation



Error Detection

Error detection at the data link layer ensures that data received over a physical link is identical to what was sent.

Error detection have many methods like :

1. CRC (Cyclic Redundancy Check)
2. Parity Check (VRC)
3. Checksum
4. LRC (Longitudinal Redundancy Check)

CRC

CRC is method adding 8 to 32 check bits of error-detection code to a block of data .

Advantages :

1. Wide Applicability
2. high reliability in validating data integrity

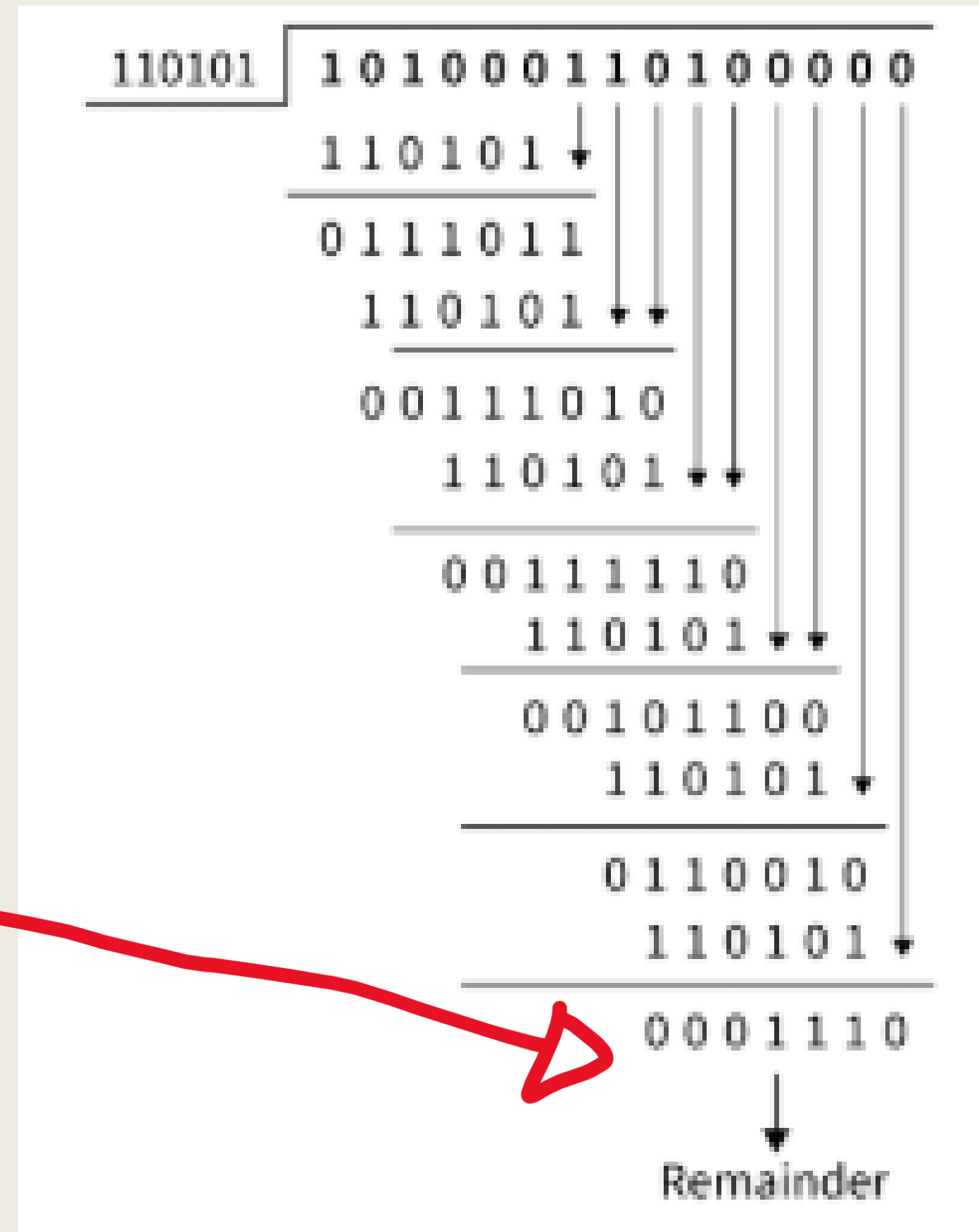
Disadvantages :

1. Errors cannot be corrected, and some errors may be forget depending on the chosen polynomial.

CRC

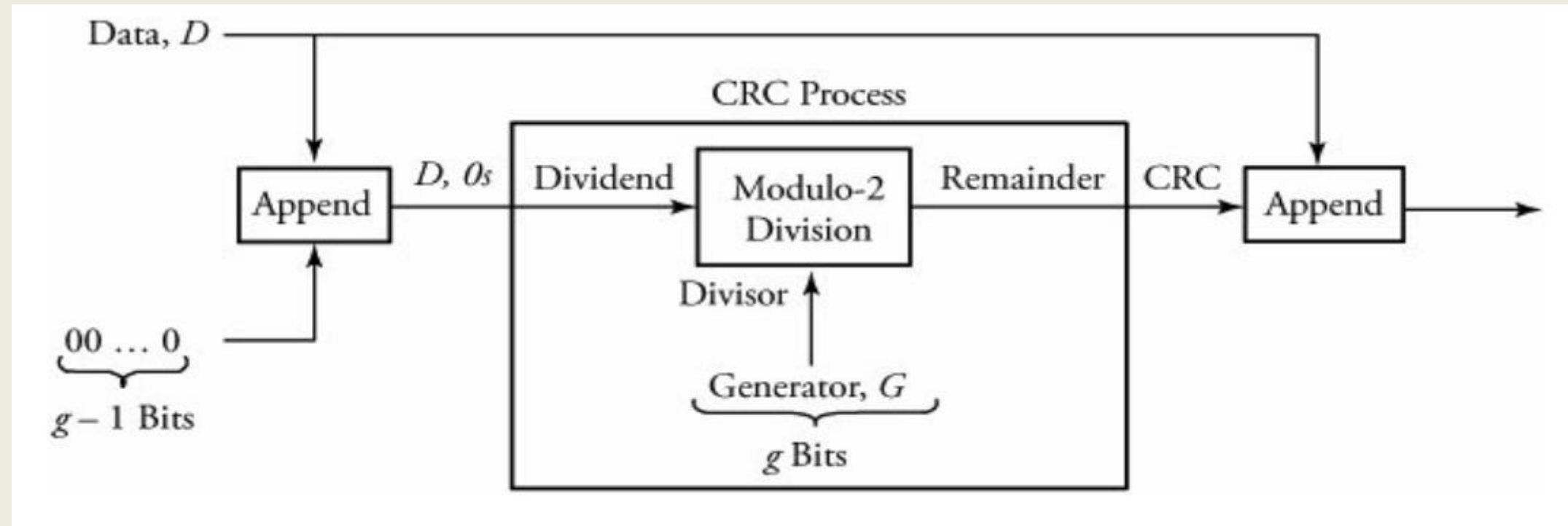
If result zeros it's correct

But when any 1 found in result ,
Error Detected.

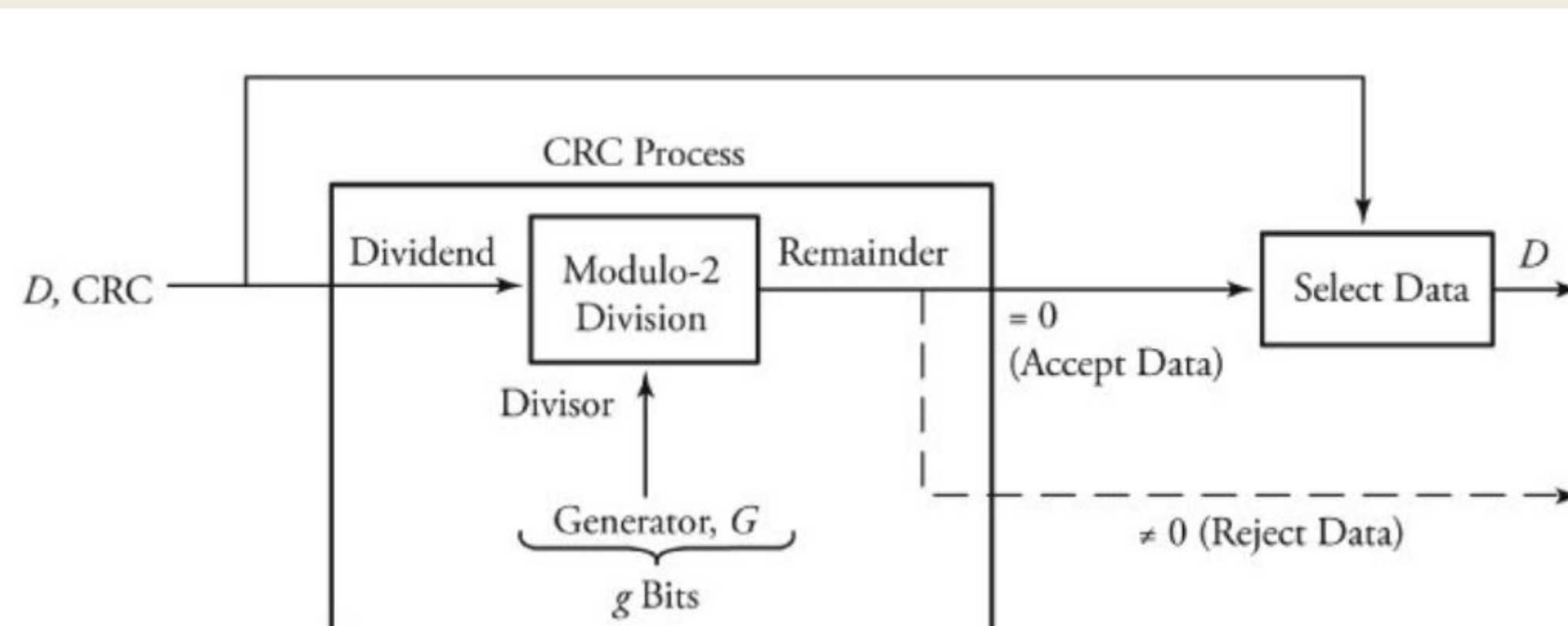


CRC

At transmitter

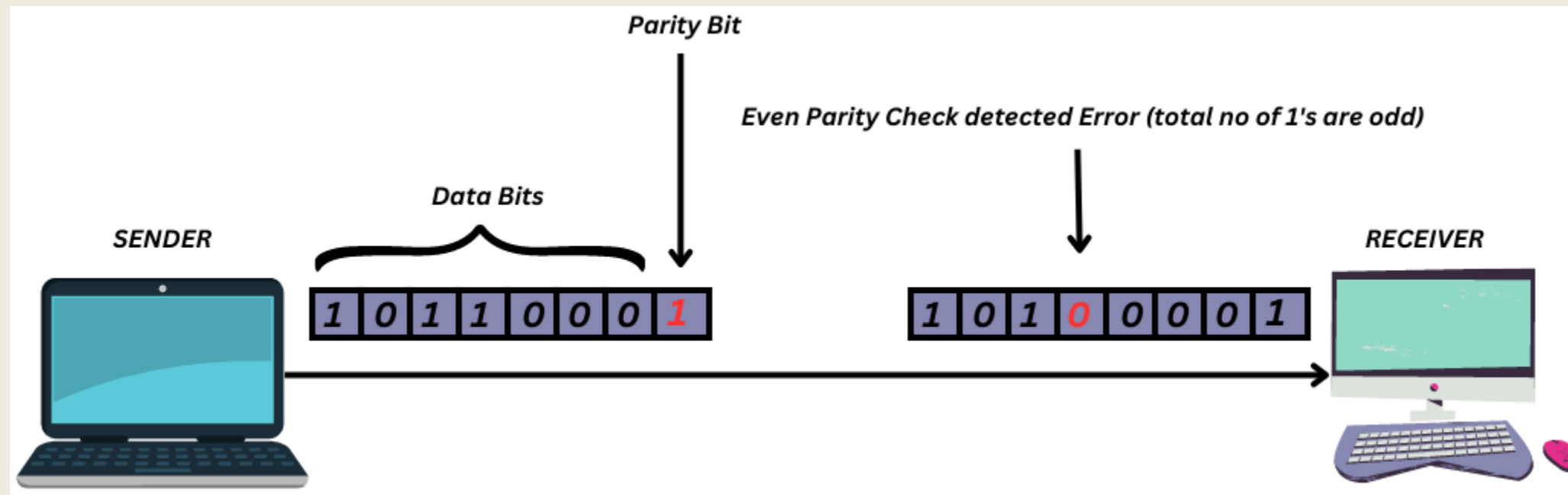


At receiver



Parity Check

basic error-detection technique used in digital communications to ensure data integrity by adding an extra bit, called a parity bit .



Parity Check

Advantages :

1. Simple to Use

Disadvantages :

1. Cannot Detect Even Errors
2. No Error correction.
3. Limited Scope to detect errors

Checksum

Error detection method using algorithm to find and detect errors with unique digital fingerprint to verify data integrity.

This method using sum and sum complement “checksum” to detect errors

10101001	subunit 1
00111001	subunit 2
00011101	checksum
11111111	sum
00000000	sum's complement

LRC (2D Parity Check)

error detection method which is used by upper layers to detect error in data.

The other name for LRC is 2-D parity check.

To detect an error, a redundant bit is added to the whole block after addition this block is transmitted to receiver side.

LRC

Advantages :

1. Simple to developing and management
2. Good for multiple and even errors

Disadvantages :

1. Cannot Detect Even Errors
2. If two bits in one data unit are damaged, and the same bits in the same position are damaged in another data unit, the system will not be able to detect this type of error.

Data flow control

Communication systems must use flow-control techniques on their transmission links to guarantee that a transmitter does not overwhelm a receiver with data.

Also we using data flow to ensures and prevent noise and interference.

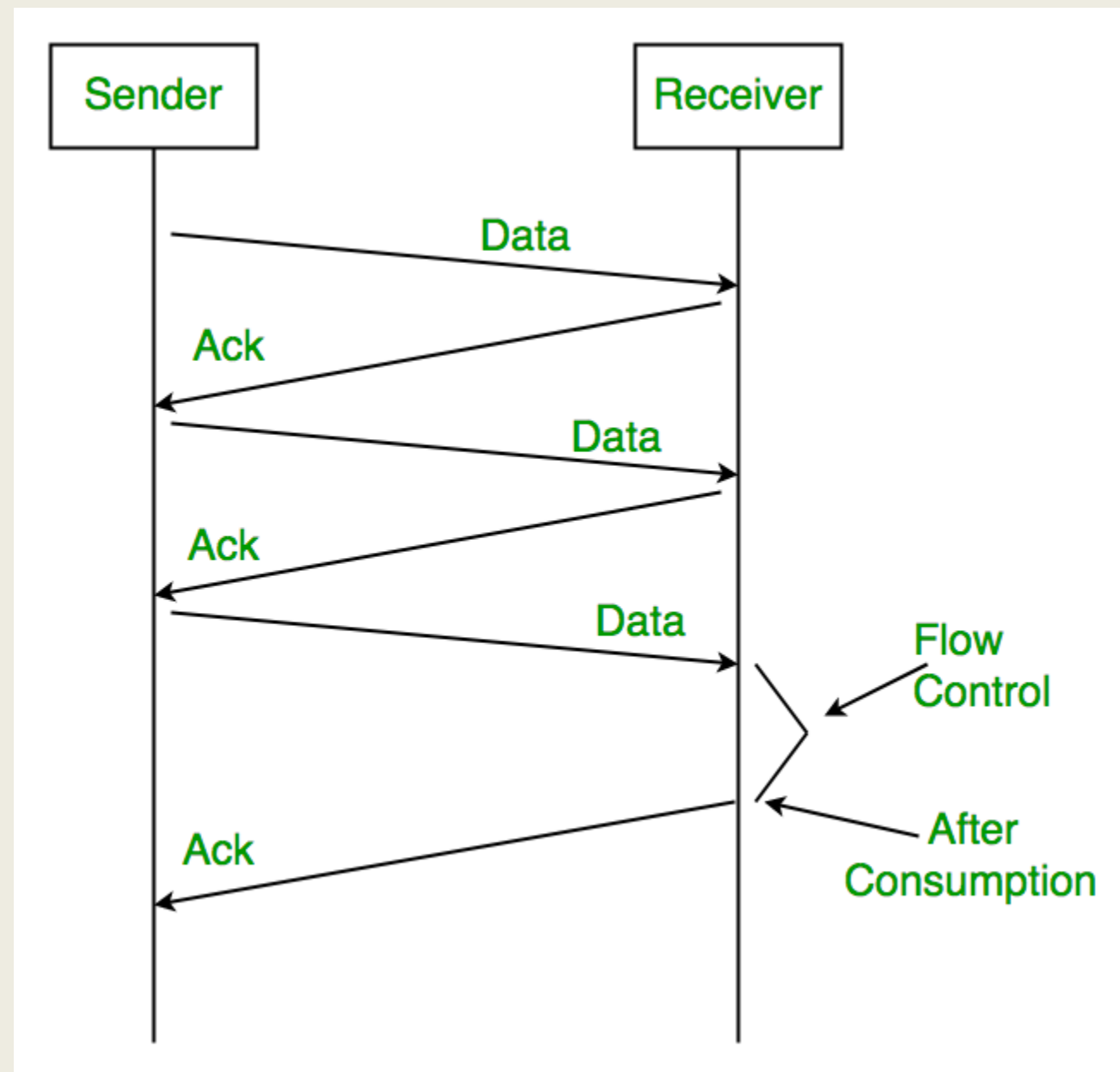
Can be used two main methods to data flow control :

- Stop and wait
- Sliding window

Stop and wait

reliable data link layer protocol in which the sender transmits one frame at a time and waits for an acknowledgment (ACK) before sending the next frame.

sender transmits only one data packet at a time , Then sender sends the next packet only after receiving an acknowledgment (ACK) for the previously sent packet.



Stop and wait

Advantages :

1. Simple to use

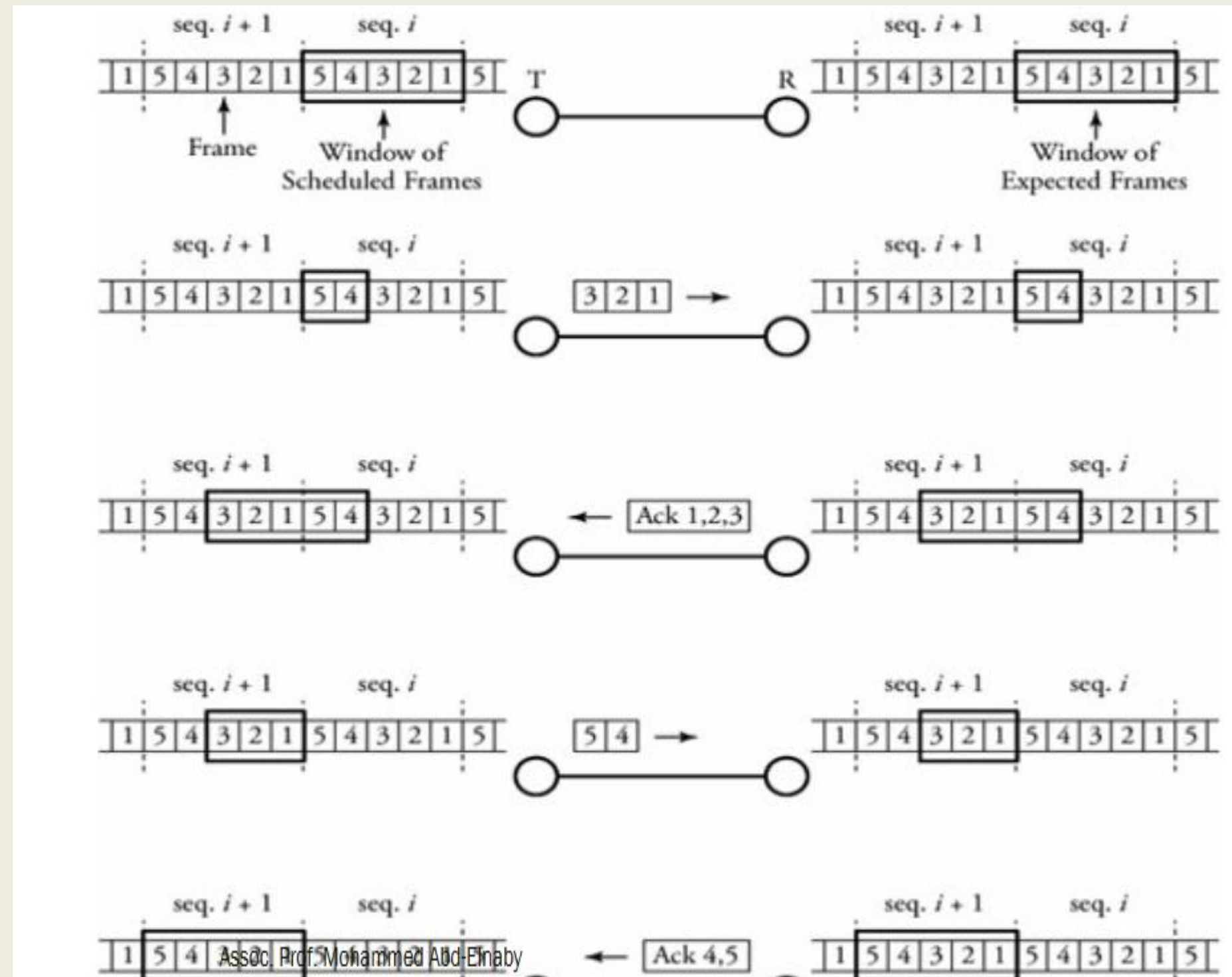
Disadvantages :

1. Data can be loss
2. Many problems such as lost acknowledgment or Delayed Acknowledgement/Data.

Sliding Window

method used to solve problems that involve subarray or substring or window.

It allows sending multiple frames (determined by window size) at once.



Sliding Window

There are two types of sliding window :

1. Fixed Size Sliding Window
2. Variable Size Sliding Window

Advantages :

1. High Efficiency and speed
2. Full Duplex support

Disadvantages :

1. Sequence Number Limitations
2. Buffer Storage Requirements

Conclusion

The data link layer is one of the most important layers in the world of networks, whether in the OSI model or under the the network layer in the TCP/IP model, because it is the layer in which most data errors occur, as well as the methods for identifying, determining, and correcting these errors and their flow.

Vocab

Data link

Data errors

Echo

Jitter

Spike

Attenuation

Noise

Error detection

Cyclic Redundancy Check

Longitudinal Redundancy Check

Checksum

Stop and wait

Sliding window

Fixed

Variable

Even

Odd

Flow Control

Data Flow

Delay

Sources

- Taif University – Computer Networks ch.4
- Wikipedia – Cyclic Redundancy Check
- Geeksforgeeks – Error Detection in Computer Networks
- Networking for Dummies Book
- Data communications and networking book

Thanks for listen